All commutators needed in CT theory

generated by `generate_commutators.py`

The commutators of the transcorrelated (CT) expansion, generated symbolically with sympy's `Commutator` and kept in nested-operator form. They are *not* expanded into concrete many-body equations; the expansion through the generalized Wick theorem and the cumulant expansion is done separately.

The truncation is by perturbation order (Møller–Plesset counting $\hat{F} \sim O(1)$, $\hat{V} \sim O(\lambda)$, $\hat{T} \sim O(\lambda)$), keeping the expansion correct through $O(\lambda^2)$: the full first-order term $[\hat{H}, \hat{A}]$ ($O(\lambda) + O(\lambda^2)$) and the Fock half of the second-order term, $\frac{1}{2}[[\hat{F}, \hat{A}], \hat{A}]$ ($O(\lambda^2)$). The valence double commutator $\frac{1}{2}[[\hat{V}, \hat{A}], \hat{A}]$ is $O(\lambda^3)$ and is not generated.

1 Operators and generator

Convention of `ct_F_A.tex`, `ct_V_A.tex` and `ct_f_a_a.pdf`: crossed Fock order, excitation operators, general indices over $\{i\} \cup \{a\}$:

$$\hat{F} = f^q_p \, \hat{E}^p_q, \qquad \hat{V} = \tfrac{1}{2} g^{pq}_{rs} \, \hat{E}^{rs}_{pq}, \tag{1}$$

$$\hat{T} = \hat{T}_1 + \hat{T}_2, \qquad \hat{A} = \hat{T} - \hat{T}^\dagger, \tag{2}$$

$$\hat{T}_1 = t^a_i \, \hat{E}^a_i, \qquad \hat{T}_2 = \tfrac{1}{2} t^{a\beta}_{ij} \, \hat{E}^{a\beta}_{ij}, \tag{3}$$

with $\hat{F}$ and $\hat{V}$ Hermitian and $\hat{A}$ anti-Hermitian. The blocks of the first-order commutators are

$$\hat{C}_1 = \left[\hat{F}, \hat{T}_1\right], \qquad \hat{C}_2 = \left[\hat{F}, \hat{T}_2\right], \tag{4}$$

$$\hat{D}_1 = \left[\hat{V}, \hat{T}_1\right], \qquad \hat{D}_2 = \left[\hat{V}, \hat{T}_2\right], \tag{5}$$

so that $\hat{C} = [\hat{F}, \hat{A}] = \hat{C}_1 + \hat{C}_2 + \hat{C}_1^\dagger + \hat{C}_2^\dagger$ and $\hat{D} = [\hat{V}, \hat{A}] = \hat{D}_1 + \hat{D}_2 + \hat{D}_1^\dagger + \hat{D}_2^\dagger$.

2 First-order commutators

Since $\hat{A} = \hat{T} - \hat{T}^\dagger$ and $[\hat{X}, \hat{T}]^\dagger = -[\hat{X}, \hat{T}^\dagger]$ for Hermitian $\hat{X}$, only the two $\hat{T}$ blocks have to be computed for each part of the Hamiltonian:

$$\left[\hat{F}, \hat{A}\right] = \left[\hat{F}, \hat{T}_1\right] + \left[\hat{F}, \hat{T}_2\right] + \text{h.c.}, \tag{6}$$

$$\left[\hat{V}, \hat{A}\right] = \left[\hat{V}, \hat{T}_1\right] + \left[\hat{V}, \hat{T}_2\right] + \text{h.c.}. \tag{7}$$

3 Second-order commutators

At $O(\lambda^2)$ the second-order term keeps only the Fock half, $\frac{1}{2}[[\hat{F},\hat{A}],\hat{A}]$; the valence half $\frac{1}{2}[[\hat{V},\hat{A}],\hat{A}] \sim O(\lambda^3)$ is dropped. $\hat{A}$ anti-Hermitian makes $\hat{C} = [\hat{F},\hat{A}]$ Hermitian, so that $[\hat{C},\hat{T}]^\dagger = -[\hat{C},\hat{T}^\dagger]$ and

$$\frac{1}{2}\left[\left[\hat{F},\,\hat{A}\right],\,\hat{A}\right] = \tfrac{1}{2}\left[\hat{C},\,\hat{T}\right] + \text{h.c.}, \tag{8}$$

Expanding $[\hat{C},\hat{T}] = [\hat{C}_1,\hat{T}] + [\hat{C}_2,\hat{T}] + [\hat{C}_1^\dagger,\hat{T}] + [\hat{C}_2^\dagger,\hat{T}]$ and each $[\,\cdot\,,\hat{T}] = [\,\cdot\,,\hat{T}_1] + [\,\cdot\,,\hat{T}_2]$ gives the eight blocks

$$\left[\left[\hat{F},\,\hat{T}_1\right],\,\hat{T}_1\right] \tag{9}$$
$$\left[\left[\hat{F},\,\hat{T}_1\right],\,\hat{T}_2\right] \tag{10}$$
$$\left[\left[\hat{F},\,\hat{T}_2\right],\,\hat{T}_1\right] \tag{11}$$
$$\left[\left[\hat{F},\,\hat{T}_2\right],\,\hat{T}_2\right] \tag{12}$$
$$\left[\left[\hat{F},\,\hat{T}_1\right]^\dagger,\,\hat{T}_1\right] \tag{13}$$
$$\left[\left[\hat{F},\,\hat{T}_1\right]^\dagger,\,\hat{T}_2\right] \tag{14}$$
$$\left[\left[\hat{F},\,\hat{T}_2\right]^\dagger,\,\hat{T}_1\right] \tag{15}$$
$$\left[\left[\hat{F},\,\hat{T}_2\right]^\dagger,\,\hat{T}_2\right] \tag{16}$$

The mixed nested commutators are different operators, $\left[\left[\hat{F},\,\hat{T}_1\right],\,\hat{T}_2\right] \neq \left[\left[\hat{F},\,\hat{T}_2\right],\,\hat{T}_1\right]$, since $\hat{T}_1\hat{T}_2 - \hat{T}_2\hat{T}_1$ does not vanish; the Jacobi shortcut is not available and both mixed blocks are kept.

4 Complete list

The commutators that remain to be evaluated (each by one or two applications of the $(1,1)$, $(1,2)$, $(2,1)$ or $(2,2)$ rule):

First order (4 blocks + h.c.)

$$\left[\hat{F},\,\hat{T}_1\right] \tag{17}$$
$$\left[\hat{F},\,\hat{T}_2\right] \tag{18}$$
$$\left[\hat{V},\,\hat{T}_1\right] \tag{19}$$
$$\left[\hat{V},\,\hat{T}_2\right] \tag{20}$$

Second order (8 blocks + h.c.)

$$\left[\left[\hat{F},\hat{T}_1\right],\hat{T}_1\right] \tag{21}$$

$$\left[\left[\hat{F},\hat{T}_1\right],\hat{T}_2\right] \tag{22}$$

$$\left[\left[\hat{F},\hat{T}_2\right],\hat{T}_1\right] \tag{23}$$

$$\left[\left[\hat{F},\hat{T}_2\right],\hat{T}_2\right] \tag{24}$$

$$\left[\left[\hat{F},\hat{T}_1\right]^\dagger,\hat{T}_1\right] \tag{25}$$

$$\left[\left[\hat{F},\hat{T}_1\right]^\dagger,\hat{T}_2\right] \tag{26}$$

$$\left[\left[\hat{F},\hat{T}_2\right]^\dagger,\hat{T}_1\right] \tag{27}$$

$$\left[\left[\hat{F},\hat{T}_2\right]^\dagger,\hat{T}_2\right] \tag{28}$$